

CHAPTER#05"LIQUIDS"**DEFINITION:**

*"All those substances which possess definite volume but not definite shape are called Liquids."*

They take the shape of container in which they are contained.

**KINETIC THEORY OF LIQUIDS:**

This theory is based on the following assumptions:

- ✓ The particles of a liquid are very close to each other due to which a liquid has fixed volume.
- ✓ The particles in a liquid are free to move so they have no definite shape.
- ✓ During the motion these molecules collide with each other and with the walls of the container.
- ✓ These molecules possess kinetic energy which is directly proportional to its absolute temperature.

**"PROPERTIES OF LIQUIDS"**

Some of the properties of liquids are given below:

**DIFFUSIBILITY****DEFINITION:**

*"The mixing of two substances so as to form a homogeneous mixture is called diffusion."*

**BRIEF EXPLANATION:**

According to kinetic molecular theory, the liquid molecules are in cluster form and are present closer to each other. A strong attractive force binds them. Their particles are fairly free to move. Hence the molecules of liquids can diffuse into each other but the rate of diffusion in liquids is lower than the gases.

**COMPRESSIBILITY****DEFINITION:**

*"Liquids are normally incompressible. However at very high pressure the volume of liquid is reduced very slightly."*

**KMT EXPLANATION:**

According to kinetic molecular theory, the intermolecular spaces in liquids are lesser than those of gases molecules. Hence they cannot be compressed easily. However a high pressure is required to compress a liquid to a very little extent.

**EXPANSION**

Some of the liquids on heating show expansion i.e. increase in their volumes. Due to increase in temperature the kinetic energy of molecules increases due to which they move apart.

**CONTRACTION**

On cooling liquids show contraction i.e. decrease in their volume. It is due to cooling process where thermal energy of the molecules is removed which causes decrease in kinetic energy and decrease in intermolecular spacing.

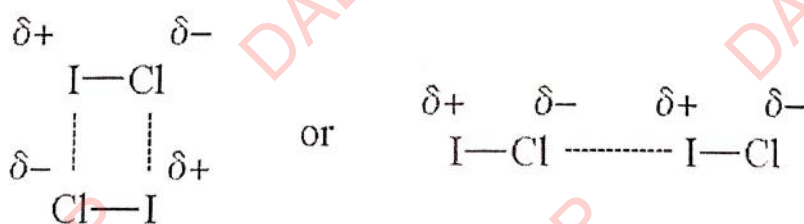
**"INTER-MOLECULAR FORCES IN LIQUIDS"**

It is defined as,

*"The attraction and repulsions between atoms, molecules, and surfaces, as well as other intermolecular forces are called as Vander-wall force."*

**➤ DIPOLE-DIPOLE INTERACTION:**

Dipole-dipole forces are attractive forces between the positive end of one polar molecule and the negative end of another polar molecule.

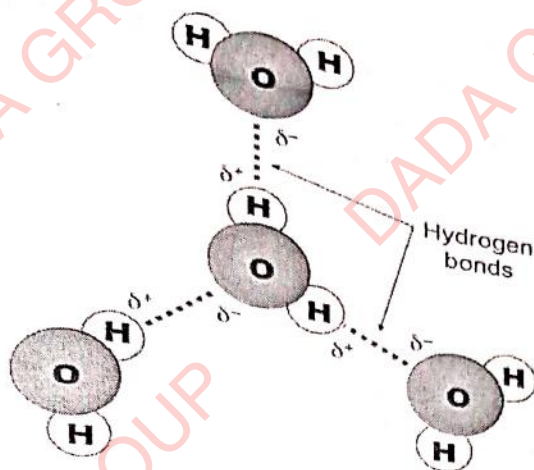


The figures show two arrangements of polar iodine monochloride (ICl) molecules that give rise to dipole-dipole attractions.

**➤ HYDROGEN BONDING:**

Hydrogen bonding is a special type of dipole-dipole attraction between molecules, not a covalent bond to a hydrogen atom. It results from the attractive force between a hydrogen atom covalently bonded to a very electronegative atom.

It is represented by dotted lines (.....). It is strongest in all inter-molecular forces however, much weaker than an ordinary covalent bond.



## "PHYSICAL PROPERTIES OF LIQUIDS"

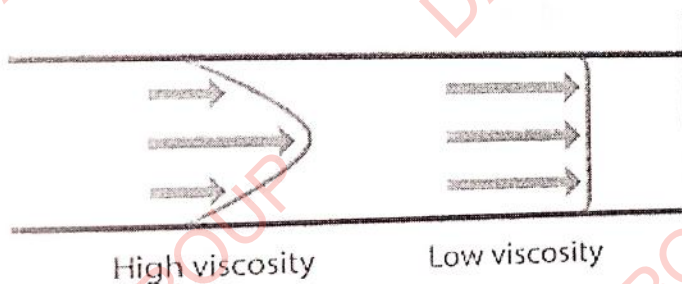
### VISCOSITY

#### DEFINITION:

*"The internal resistance in flow of a liquid is called viscosity."*

#### EXPLANATION:

The internal resistance of flow of a liquid is due to the internal friction between the two layers of the liquid. This could easily be understood by considering the liquid in a tube, consists of concentric layers. When this liquid is allowed to flow, the layers, which are in contact with the wall of the tube, are stationary while the layer at the center has high velocity. The intermediate layers move with medium velocities. Hence each layer exerts a drag to the next layer which causes the liquid to flow.



#### FACTORS EFFECTING VISCOSITY OF LIQUIDS:

Following are the factors which affect the viscosity of the liquids.

##### ✓ SIZE OF MOLECULES:

The viscosity of a liquid depends upon the size of the liquid molecules. If the size of the molecules is bigger then the viscosity of liquid is greater.

✓ **SHAPE OF MOLECULES:**

An irregular shape of molecules offers more resistance than the molecules of regular shape. Thus the non-linear molecules have greater viscosity than the linear ones.

✓ **INTERMOLECULAR ATTRACTION:**

The liquids having strong molecular attractions are more viscous than the liquids with weaker molecular attractions.

✓ **TEMPERATURE:**

Viscosity decreases with the increase in temperature because at high temperature, the attractive forces between the molecules become weak.

✓ **MEASUREMENT OF VISCOSITY:**

The instrument which is used to measure viscosity is called "OSTWALD VISCOMETER".

**REPRESENTATION:**

Viscosity is represented by  $\eta$  (eta).

**UNITS:**

- ✓ Poise (1 poise = 1 g/cm )
- ✓ Centipoises
- ✓ Mill poise
- ✓ N.sec/m<sup>2</sup>. (S.I. unit)

**SURFACE TENSION**

**DEFINITION:**

*"The force acting perpendicular to the surface of a liquid is called surface tension."*

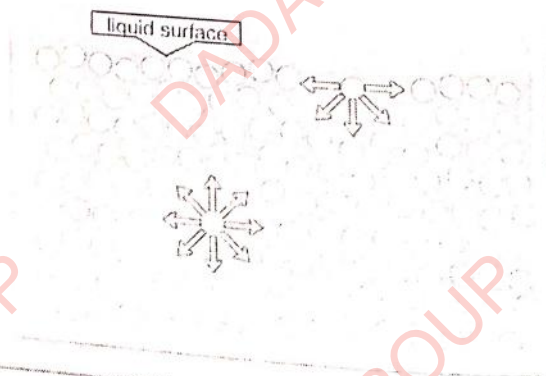
OR

*"The energy per unit surface area of a liquid is called surface tension."*

**EXPLANATION:**

The molecules of liquid are pulled equally below its surface from all sides but the molecules at the surface are attracted by many molecules below the surface and not attracted by molecules above the surface.

Therefore the molecules at the surface feel a net attraction inward. This attraction creates surface tension and the surface of liquid behaves like a stretched membrane.



- UNITS:
- ✓ Dynes/cm.
  - ✓ Erg/cm<sup>2</sup>.
  - ✓ N/m (S.I. unit)

### FACTORS AFFECTING SURFACE TENSION:

Following are the factors which affect the surface tension of liquids.

#### ✓ INTERMOLECULAR FORCE:

The greater the intermolecular force the greater will be the surface tension.

#### ✓ EFFECT OF TEMPERATURE:

On increasing temperature the force of attraction between the molecules decreases. Hence the surface tension of the liquid decreases.

#### ✓ EFFECT OF HYDROGEN BONDING:

The greater the hydrogen bonding among the molecules, the more is the surface tension. Water has high surface tension due to Hydrogen bonding.

#### ✓ PRESENCE OF IMPURITIES:

By adding the impurities the surface tension decreases. It means the surface tension of pure liquid is always higher than impure liquid.

### APPLICATIONS OF SURFACE TENSION:

- ✓ A tissue paper absorbs ink due to surface tension.
- ✓ Capillary action takes place due to surface tension.

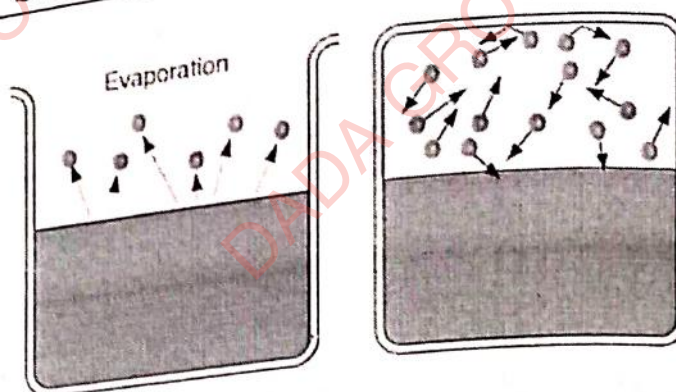
## VAPOUR PRESSURE

### DEFINITION:

*"The pressure exerted by vapors of a liquid when they are in equilibrium with liquid phase is called Vapor pressure."*

### EXPLANATION:

When a liquid is heated in an open container some of the vapours convert back to the liquid state but most of the vapours escape into surroundings and equilibrium is not established between liquid and vapour phase. When a liquid is heated in closed container about all the vapours convert back to the liquid state hence condensation takes place and equilibrium is established between evaporation and condensation. When liquid and vapour phases are at equilibrium to each other, at that time the pressure exerted by the vapour is called Vapour Pressure.

**UNITS:**

1. Millimeter of Hg
2. Atmosphere
3. Torr
4. Newton/m<sup>2</sup> or Pa

**FACTORS EFFECTING VAPOUR PRESSURE: ,**

Following are the factors which affect the surface tension of liquids.

**✓ NATURE OF LIQUID:**

Vapor pressure of liquid depends upon nature of liquid. Liquids having low boiling point exert more pressure at a given temperature.

**✓ TEMPERATURE:**

Vapor pressure of liquid also depends upon temperature. It increases with increase in temperature due to increase in kinetic energy.

**✓ INTERMOLECULAR FORCES:**

The liquids having strong molecular attractions have lower vapor pressure than the liquids with weaker molecular attractions.

**✓ PRESENCE OF IMPURITIES:**

By adding the impurities the vapor pressure decreases. It means the vapor pressure of pure liquid is always higher than impure liquid.

**EVAPORATION****DEFINITION:**

*"The spontaneous escaping of liquid molecules from the surface of liquid without boiling is called Evaporation."*

**K.M.T EXPLANATION:**

Evaporation occurs when high energy molecules at the surface of liquid break away from their neighbors and escape into the gaseous phase. In other words, the molecules of liquid collide with

each other; due to these collisions some of the molecules acquire higher energy so they escape into the air from the surface in the form of vapors.

### FACTORS AFFECTING EVAPORATION:

#### ✓ SURFACE AREA:

Greater the surface area of liquid greater would be the evaporation.

#### ✓ TEMPERATURE:

Evaporation occurs at all the temperature but increase in the temperature of liquid increases the evaporation.

#### ✓ NATURE OF LIQUID:

Different liquids evaporate at different rates. This is because of the extent of intermolecular forces. Liquids which evaporate more quickly are called as Volatile liquids.

## BOILING POINT

### DEFINITION:

*"The temperature at which vapors pressure of liquid become equal to atmospheric pressure or any other external pressure is called boiling point".*

### EXPLANATION:

When a liquid is heated the rate of evaporation of the molecules also increases, when the pressure of liquid become equal to the atmospheric pressure the liquid starts boiling and this temperature is known as boiling point of liquid. Boiling point of the liquid decreases with the increase in external pressure and vice versa.

## CAPILLARY ACTION

### DEFINITION:

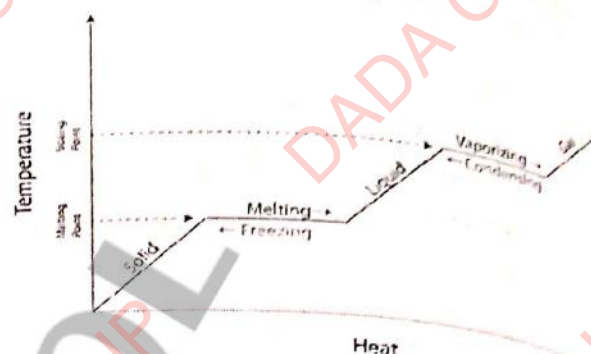
*"The spontaneous rise of a liquid in a capillary tube is called capillary action."*

### EXPLANATION:

Capillary action is due to surface tension. Capillary is a glass tube having very small inner diameter. When a capillary tube immersed in a liquid, the liquid creeps up in the tube until a balance is reached between adhesive forces and weight of liquid in Capillary. The smaller the bore, higher the liquid climb. This is known as Capillary Action.

**"ENERGETIC OF PHASE CHANGE"**

Whenever matter undergoes a physical or a chemical change, it is always accompanied by an energy change in the form of heat. This change in energy is the quantitative measure of the difference in the strength of intermolecular forces.

**MOLAR HEAT OF FUSION:**

It is the amount of heat absorbed by one mole of a solid when it melts into liquid form at its melting point at one atmospheric pressure.

**MOLAR HEAT OF VAPORIZATION:**

The amount of heat absorbed when one mole of a liquid is changed to its vapour at its boiling point at one atmospheric pressure is called molar heat of vaporization.

**MOLAR HEAT OF SUBLIMATION:**

The amount of heat absorbed when one mole of a solid sublimates to give vapour at a particular temperature at one atmospheric pressure is called molar heat of sublimation.

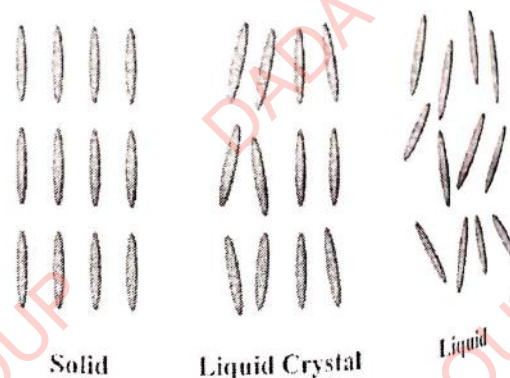
**"LIQUID CRYSTALS"****INTRODUCTION:**

A liquid crystal is a state of matter between liquid and solid. It was first discovered by Friedrich Reintzer (1888), while he was examining physico-chemical properties of different derivatives of cholesterol. He observed that cholesteryl-Benzoate did not melt the same way as other compounds, and it has two melting points. At  $145.5^{\circ}\text{C}$  it first melted into a cloudy liquid and then at  $178.5^{\circ}\text{C}$  it melted again and cloudy liquid became clear and this phenomenon was reversible.

**PROPERTIES OF LIQUID CRYSTAL:**

Liquid crystals are usually organic in nature or may be sometimes mixture of both organic and inorganic materials. Liquid crystals exist both naturally or they can be synthetically developed.

They have ability to flow like liquid but the molecular arrangement similar to crystalline solids. Liquid crystal have optical properties as found in crystalline so but exhibit anisotropy.



Solid

Liquid Crystal

Liquid